

Project Title: Large Number Arithmetic Using Doubly Linked List

This project focuses on the design and implementation of a system that performs **arithmetic operations on very large integers using a doubly linked list data structure**. Since such numbers may exceed the storage capacity of built-in numeric data types, each digit is stored individually in a node of the doubly linked list. **The project supports arithmetic operations such as addition, subtraction, multiplication and division.**

Goal: Through this project, students are expected to understand how doubly linked lists can be applied in practical computational problems involving large numbers, while also strengthening their knowledge of linked list traversal, dynamic memory management, and digit-by-digit arithmetic processing.

Below are some examples.

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Example 1:

Input:
m = 123456789123456789123456789123456789123456789
n = 456789123456789123456789123456789123456789

Output:
addition = 123913578246913578246913578246913578246913578
subtraction = 1230000000000000000000000000000000000000000
multiplication =
563937184884934839205932493526930147847927802168925...303510198119182004648682028105472051562262
0750190521
division = 270.27085975512599935528

Example 2:

Input:
m = 55
n = 2

Output:
addition = 57
subtraction = 53
multiplication = 110
division = 27.5
```

Approach: The large number is first divided into individual digits and stored in a doubly linked list. Addition and subtraction are performed digit by digit following the standard arithmetic method, including the handling of carries and borrows. These basic operations are then used to implement multiplication and division. Multiplication is carried out by multiplying each digit step by step, followed by shifting and adding intermediate results, while division is performed by identifying suitable multiples of the divisor and subtracting them from the dividend progressively. Finally, the computed results are presented through a display function.